

# ISO-1 weak isogeny classes: a norm-one conductor theorem and trace census

Research evidence report

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**Contribution.** Over every odd extension of  $\mathbb{F}_q$ , with  $q \equiv 1 \pmod{4}$ , the norm-one weak family has a Legendre twist whose rational 2-isogeny target carries full rational 4-torsion. Hence  $t \equiv \pm(q^n + 1) \pmod{16}$ , and every ordinary weak class has  $4 \mid f_\pi$ . An exact  $p = 7$  census proves that the converse fails: the existing full-2 class  $t = \pm 610$  has conductor 72 and no weak representative. Absolute Frobenius and inversion reduce the cubic census to exactly  $(p^4 + 3p^2 + 8)/12$  point-count calls. The [standalone theorem note](#) gives the generalized proof, exceptional-orbit classification, and counterexample certificate.

**Requested scope.** Base primes  $p = 37$  through about 200,  $q = p^2$ , and curves over  $\mathbb{F}_{q^3} = \mathbb{F}_{p^6}$ : label every full-2-torsion isogeny class and find a criterion for weak-class reach. This round completed every-trace censuses at  $p = 37, 41, 43, 47, 53$  and diagnostic censuses at  $p = 11, 13, 17$ . The validated  $p = 53$  census adds 72,540 weak ordinary rows and 494 high-depth zero rows. The worker has started  $p = 59$  and retains all remaining primes through  $p = 199$ .

**Method.** A weak representative has  $y^2 = x(x - \alpha)(x - \sigma\alpha)$ , with  $\sigma$  the  $p^2$ -Frobenius. After translation and  $\mathbb{F}_{p^2}$  square scaling there are  $2q^2 + 2q$  normalized representatives. The  $p = 37$  run evaluated one representative per quadratic-twist pair and derived the opposite trace. Trace assignment uses a randomized two-point baby-step/giant-step counter. Its labels are computational evidence, not independent certificates of every cardinality. Full and twist-derived CSVs agreed exactly at  $p = 13$ ; two false individual assignments at  $p = 7$  were found by exact square-table counting. The new  $p = 7$  census instead counts every field element for all 213 absolute orbits; its complete weighted trace distribution agrees with independent PARI/GP counts of all those orbits. Independent PARI/GP `ellcard` checks placed 100, 3,952, 4,043, and 4,200 distinct norm-one Legendre traces at  $p = 37, 41, 43, 47$  in the respective positive sets (5,000 samples each at  $p = 41, 43, 47$ ). At  $p = 53$ , all 5,000 controls (4,404 distinct absolute traces) are positive.

| $p$       | $q$          | weak reps         | ordinary $t$   | weak $t$      | depth-1 weak    | depth- $\geq 2$ zero |
|-----------|--------------|-------------------|----------------|---------------|-----------------|----------------------|
| 7         | 49           | 4,900             | 294            | 126           | 0/148           | 20/146               |
| 11        | 121          | 29,524            | 1,210          | 542           | 0/606           | 62/604               |
| 13        | 169          | 57,460            | 2,028          | 928           | 0/1,014         | 86/1,014             |
| 17        | 289          | 167,620           | 4,624          | 2,198         | 0/2,312         | 114/2,312            |
| <b>37</b> | <b>1,369</b> | <b>3,751,060</b>  | <b>49,284</b>  | <b>24,352</b> | <b>0/24,642</b> | <b>290/24,642</b>    |
| <b>41</b> | <b>1,681</b> | <b>5,654,884</b>  | <b>67,240</b>  | <b>33,224</b> | <b>0/33,620</b> | <b>396/33,620</b>    |
| <b>43</b> | <b>1,849</b> | <b>6,841,300</b>  | <b>77,658</b>  | <b>38,452</b> | <b>0/38,830</b> | <b>376/38,828</b>    |
| <b>47</b> | <b>2,209</b> | <b>9,763,780</b>  | <b>101,614</b> | <b>50,382</b> | <b>0/50,808</b> | <b>424/50,806</b>    |
| <b>53</b> | <b>2,809</b> | <b>15,786,580</b> | <b>146,068</b> | <b>72,540</b> | <b>0/73,034</b> | <b>494/73,034</b>    |

At  $p = 37$  the run took 1,106.807 seconds with four Rayon threads and counted 563,856,133,786  $\mathbb{F}_p$  multiplications. The 50,654 CSV rows cover every candidate  $t \equiv 2 \pmod{4}$  in the Hasse interval; 49,284 are ordinary. Of 4,000 sampled random full-2-torsion curves, 2,433 were in an observed weak class: 0.6083, Wilson 95% interval [0.5930, 0.6233]. The zero-label share at  $p = 37$  is 50.59% of ordinary trace rows but 39.18% of sampled curves; these use different denominators. The  $p = 41, 43, 47$  sampled reach fractions are 0.6253, 0.6195, and 0.6320 (4,000 curves each; Wilson 95% intervals [0.6101, 0.6401], [0.6043, 0.6344], and [0.6169, 0.6468]). At  $p = 47$  the special size-two orbit yields the two nonordinary Hasse-boundary traces  $t = \pm 2 \cdot 47^3$  (two representatives each), independently confirmed with PARI/GP.

The  $p = 53$  run uses 658,243 point counts, weighted to all 15,786,580 representatives, and writes 148,878 Hasse rows. It records 313,857,554,508 charged  $\mathbb{F}_p$  multiplications and 2,057.706 wall seconds with four workers on the contended host. Its field tower is  $u^2 = 2$ ,  $\theta^3 = 1 + u$ , with  $\theta^q = (26 + 24u)\theta$ . Its 4,000-curve sample has 2,539 hits: 0.6348, Wilson 95% interval [0.6197, 0.6495].

**Proof of the necessary obstruction.** Put  $K = \mathbb{F}_{q^n}$  and  $Q = q^n$ , where  $q \equiv 1 \pmod{4}$  is odd and  $n \geq 3$  is odd. On a weak model  $y^2 = x(x - \alpha)(x - \alpha^q)$ , with  $\alpha \notin \mathbb{F}_q$ , put  $\lambda = \alpha^q/\alpha$ . Then  $N_{K/\mathbb{F}_q}(\lambda) = 1$ , so  $\lambda$  has odd order dividing  $1 + q + \cdots + q^{n-1}$ . Fourth powering is an automorphism of this subgroup; choose  $\mu^4 = \lambda$ . For  $n = 3$ , take  $\mu = \lambda^{(q^3+q+2)/4}$ . The weak curve is a quadratic twist of  $L_\lambda : y^2 = x(x-1)(x-\lambda)$ . The rational 2-isogeny at  $(0,0)$  has target

$$L'_\lambda : y^2 = x[x^2 + 2(1+\lambda)x + (1-\lambda)^2],$$

For  $x \neq 0$ , the map is

$$(x, y) \mapsto \left( x + a + \frac{b}{x}, y \left( 1 - \frac{b}{x^2} \right) \right), \quad a = -(1+\lambda), \quad b = \lambda.$$

The identity and  $(0,0)$  map to the identity. With  $U = x^2 + ax + b$ , clearing the curve equations gives  $U^2 - 2axU + (a^2 - 4b)x^2 = (x^2 - b)^2$ . The target roots are  $0, -(1+\mu^2)^2, -(1-\mu^2)^2$ . All their differences are squares: the final difference is  $\pm 4\mu^2$  and  $-1$  is a square in  $K$ . The roots are distinct since  $\lambda \neq 0, 1$ . By [Auer–Top, Lemma 2.1](#), each rational 2-torsion point has a rational half, and the halves of two independent points generate  $L'_\lambda[4]$ . Therefore  $16 \mid \#L_\lambda(K)$ , and the weak trace obeys  $t \equiv \pm(Q+1) \pmod{16}$ . For any ordinary curve with  $Q \equiv 1 \pmod{4}$ , this congruence is equivalent to  $4 \mid f_\pi$ : the integral trace and norm of  $(\varepsilon\pi - 1)/4$  give an order of conductor  $f_\pi/4$ . Conversely,  $4 \mid f_\pi$  permits a sign with that element integral, whose norm recovers the congruence. Twisting preserves  $\mathbb{Z}[\pi]$ .

Specialize to  $q = p^2$  and  $n = 3$ . Let  $t^2 - 4p^6 = f_\pi^2 D_K$ , where  $D_K$  is the fundamental CM discriminant. The measured  $v_2(f_\pi)$  is a class-wide bound, not a certified endomorphism-ring volcano level. For ordinary  $t \equiv 2 \pmod{4}$ , the arithmetic tests

$$v_2(f_\pi) \geq 2 \iff (t/2)^2 \equiv p^6 \pmod{16} \iff t/2 \equiv \pm p^3 \pmod{8}$$

are equivalent to the proved trace restriction. Indeed,  $d = (t/2)^2 - p^6$  has  $v_2(d) \geq 3$ . If  $v_2(d) = 3$ , then  $v_2(D_K) = 3$  and  $v_2(f_\pi) = 1$ ; if  $v_2(d) \geq 4$ , then  $v_2(t^2 - 4p^6) \geq 6$  and  $v_2(D_K) \in \{0, 2, 3\}$  forces  $v_2(f_\pi) \geq 2$ . There is a curve-level consequence: on the 2-isogenous neighbor with rational full 4-torsion,  $\pi - 1$  factors through the separable map [4]. Thus  $\eta = (\pi - 1)/4$  is an endomorphism, and  $\text{disc } \mathbb{Z}[\eta] = (t_L^2 - 4p^6)/16$ , where  $t_L$  is the untwisted Legendre trace. Its generated order has conductor  $f_\pi/4$ , so the neighbor's endomorphism conductor has 2-valuation at most  $v_2(f_\pi) - 2$ . Twisting preserves that order, and a degree-2 isogeny changes its 2-valuation by at most one (via the dual-isogeny inclusions of twice each endomorphism ring). Thus an ordinary weak curve satisfies  $v_2(f_{\text{End}(E)}) \leq v_2(f_\pi) - 1$ : it is never at the deepest possible 2-volcano level. Individual levels remain unmeasured. This bound also follows directly from full rational 2-torsion:  $(\pi - 1)/2$  is an endomorphism and its generated order has conductor  $f_\pi/2$ . It applies to every full-2-torsion curve in the population, so it does not further distinguish weak representatives. The extra class restriction comes from full 4-torsion on the neighbor. The accompanying identity records replay the target factorization and cleared isogeny equation at 32 recorded plus 64 additional points. Independent exact PARI/GP expansion checks both identities over the integers; the records cover those polynomial steps, while the theorem follows from the deduction above. The record IDs are IDC1h8ae7209125dfe0a2 and IDC1he02ab9a5cf019a56. The new halving and orbit-count records are IDC1hf8aba4800a6ceb5c and IDC1h1bc37ec40bcb8619. Independent generalized PARI controls check 1,100 norm-one parameters over 11 field/extension pairs and the conductor equivalence on all 476 ordinary Hasse traces at ten field sizes, with zero recorded failures.

**Certified failure of the converse.** At  $p = 7$ ,  $t = \pm 610$  satisfies  $t^2 - 4 \cdot 7^6 = 72^2(-19)$ , so  $v_2(f_\pi) = 3$ . Every-field-element counting and an independent complete PARI/GP orbit census both give zero weak representatives. Independently,  $y^2 = x^3 + 2x + 6$  over  $\mathbb{F}_{7^6}$  has trace 610 and three rational 2-torsion roots. Thus this is an existing ordinary full-2 class. The exact  $p = 7$  census has 20 ordinary high-depth zero rows in total.

**Held-out classifier.** On  $p = 37$  rows the necessary rule has 24,352 true positives, 290 false positives, no false negatives, and 24,642 true negatives (99.41% correct);  $p = 41, 43, 47$  add 396, 376, and 424 false positives, respectively. At  $p = 47$ , the necessary rule has 99.58% ordinary-row accuracy and no observed false negative. The  $p = 53$  holdout has TP 72,540, FP 494, FN 0, and TN 73,034.

**Counterexample to the proposed small invariants.** At  $p = 37$ ,  $t = -92,218$  has zero observed weak representatives and  $t = 38,854$  has 24. Both have Frobenius conductor depth 3, ramified 2, even maximal-order class number, and the same trace modulo  $2^{17}$ . Their PARI/GP class numbers are 912 and 3,168. Thus 2-splitting, conductor depth, small 2-power trace residues, and class-number parity do not classify all rows together. At  $p = 41$ , traces 136542 (zero) and 5470 (132 weak representatives) share depth 2, ramified 2, exact maximal-order class number  $h_K = 192$ , and the same trace modulo  $2^{17}$ . A near-edge pair, 133826 (zero) and 134082 (12), shares depth 3, split 2,  $h_K = 1680$ , and trace modulo 256. Class-number magnitude is associated with residual zeros, but does not yield an exact classifier even with these 2-adic features. At held-out  $p = 47$ ,  $t = 204238$  (zero) and  $t = 73166$  (96) likewise share depth 3, split 2, exact  $h_K = 336$ , and trace modulo  $2^{17}$ .

**Exact orbit reduction and audit.** Projective  $\alpha$  modulo  $\mathbb{F}_q^*$  maps bijectively to the norm-one parameters  $\lambda \neq 1$ . Their  $q$ -Frobenius conjugates and inverses have the same trace up to the quadratic twist already paired in the census. All such orbits have size six except the nontrivial cube roots of unity, which form one orbit of size two. Thus one point count per orbit suffices, reducing the present algorithm's point-count calls by nearly sixfold. At  $p = 13, 37$  the orbit CSVs match the prior direct CSVs byte for byte;  $p = 37$  needed 312,589 rather than 1,875,530 point counts. The corrected  $p = 37, 41, 43, 47$  counts obey the modulo-six constraint; at  $p = 47$  its special pair is nonordinary. This structural check is not an independent proof of every zero label. The new absolute- $p$ -Frobenius quotient uses exponents  $\pm p^j$ ,  $0 \leq j < 6$ , and has one size-2 orbit,  $(p^2 - 1)/3$  size-6 orbits, and  $(p^4 - p^2)/12$  size-12 orbits. At  $p = 13$  it reproduces the retained CSV byte for byte using 2,423 instead of 4,789 calls; all 2,423 independent GP orbit counts have the same per-trace weights. These are algebraic call reductions, independent of host timing.

**Where the residual zeros occur.** At  $p = 37$ , 234 of 290 high-depth zero rows lie in the outer fifth of  $|t|/(2p^3)$  (234/4,930 rows there, versus 6/4,926 in the inner fifth);  $p = 47$  has 392 of 424 residual zero rows in the outer fifth. This is an association, not a causal proof or an exact rule.

**Residual zeros and class-number magnitude.** The population is ordinary positive traces with  $v_2(f_\pi) \geq 2$ ; negative traces are their quadratic-twist mirrors. Native PARI/GP `qfbclassno` gives  $h_K$ . Entries are zero labels / all traces in each bin; the bins use  $q = p^2$ .

| $h_K/q$  | $p = 37$  | $p = 41$  | $p = 43$  | $p = 47$  | combined   |
|----------|-----------|-----------|-----------|-----------|------------|
| $< 1$    | 101/4,192 | 117/5,260 | 136/5,856 | 143/7,208 | 497/22,516 |
| [1, 2)   | 30/2,518  | 52/3,347  | 30/3,817  | 41/4,776  | 153/14,458 |
| [2, 4)   | 11/3,124  | 24/4,072  | 22/4,582  | 25/5,829  | 82/17,607  |
| [4, 8)   | 3/2,365   | 5/3,790   | 0/4,654   | 3/6,511   | 11/17,320  |
| $\geq 8$ | 0/122     | 0/341     | 0/505     | 0/1,079   | 0/2,047    |

These are complete observed strata, not sampled rates. The matched pairs above rule out using the listed features as an exact classifier of the recorded labels. Independent certification of every zero label remains open.

**Correction to prior evidence.** The earlier census used an  $\mathbb{F}_p$  nonsquare for the second  $\mathbb{F}_{p^2}$  square class. Every nonzero element of  $\mathbb{F}_p$  is a square in  $\mathbb{F}_{p^2}$ , so it duplicated one branch and omitted the other. The corrected code selects a genuine  $\mathbb{F}_{p^2}$  nonsquare. Use the new exact  $p = 7$  table and corrected  $p = 11, 13, 17$  tables;  $p = 23, 31$  await corrected reruns.

**Limit.** At  $p = 199$ , twist symmetry needs 1,568,278,802 point counts; the prior quotient uses 261,379,801 and the absolute quotient uses 130,696,501. The Markdown report retains the earlier resource extrapolation and its assumptions. The native worker continues sequentially through  $p = 199$ , using four workers and 5,000 independent GP controls per completed prime. It retains compressed CSVs, source hashes, and failures; a storage preflight requires one GiB plus twice a conservative raw-CSV estimate before each run. The  $p = 53$  census has passed its audit;  $p = 59$  is running and later requested primes remain unfinished. The theorem certifies absence in about half of the candidate classes and provides an exact prefilter; the certified  $p = 7$  example requires a new existence argument for a sufficient rule.

**Evidence.** The [Markdown report](#) links every CSV, exact command, identity record, sample, and [validation receipt](#). The weak model is from [Joux-Vitse, §4.1](#); the broader Legendre-family result is due to [Auer-Top](#).

## Complete observed trace strata and invariant flow

### ISO-1: trace-level weak-class census

Ordinary full-2-torsion trace candidates over  $F_{(p^6)}$ ; corrected  $F_{(p^2)}$  square classes

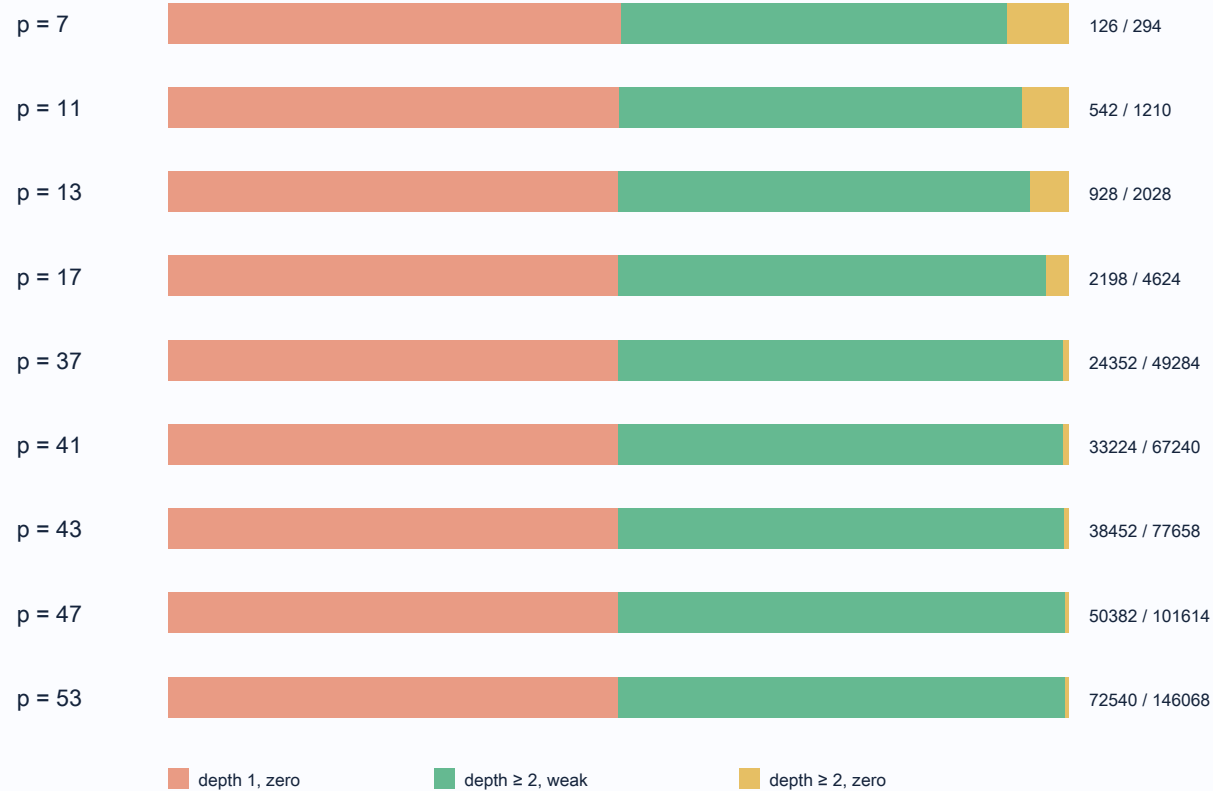
trace  $t$ ,  $t \equiv 2 \pmod{4}$   
class invariant:  $\Delta = t^2 - 4p^6$

$v_2(f_{\text{Frob}}) = 1$   
proved absent for every odd  $p$

$v_2(f_{\text{Frob}}) \geq 2$   
mostly weak; residual zero rows remain

Proof: norm-one  $\lambda \rightarrow$  fourth power  $\rightarrow$  2-isogenous full 4-torsion  $\rightarrow t \equiv \pm(p^6 + 1) \pmod{16}$

#### Class labels by stratum



Source: the nine census CSVs linked in the Markdown report, rendered by `iso1_class_census visual`. The bars count ordinary trace rows; their widths are fractions of each prime's row population. The right-hand labels give weak rows / ordinary rows. Sampling intervals apply to the separate 4,000-curve reach samples, not these census bars. The proof excludes every depth-1 class; the yellow segments retain the observed high-depth zero rows. The  $p = 7$  labels are independently exact; the larger-prime labels retain their randomized point-count uncertainty.